

Intelligent Data Analysis

DV1597

Assignment 2

Group 9

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1 Introduction

The coronavirus disease (COVID-19) pandemic has a major impact on public health systems, societies, and economies worldwide since its outbreak in 2020. European Union and European Economic Area (EU/ EEA) countries experienced several waves of infections, varying mortality rates, different vaccination strategies, and unequal pressure on healthcare system during the pandemic period. Understanding these patterns through data analysis can provide valuable insights into how different countries managed the pandemic and how factors such as geography, population size, vaccination coverage, and healthcare capacity influenced the outcomes.

The main objective of this assignment is to identify the data collected from the European Centre for Disease Prevention and Control (ECDC) by performing an exploratory and explanatory data analysis of COVID-19 and analyze important COVID-19 patterns across EU/ EEA countries using statistical analysis and visualization techniques.

2 Dataset Description and Preprocessing

Three datasets are used in this assignment are: daily numbers of COVID-19 cases and deaths, COVID-19 vaccination data, and hospital and intensive care unit (ICU) admission rates across EU/ EEA countries between 2020 and 2022. These datasets enable the investigation of infection trends, vaccination behavior, hospitalization levels, and geographical differences among countries during the pandemic.

2.1 Cases and Deaths Dataset

This dataset contains 28,729 rows and 11 columns daily reported COVID-19 cases and deaths for EU/EEA countries.

Table 1 Column Categories in the Cases and Deaths Dataset

Column Category	Column Name	Description
Time Information	- dateRep - day - month - year	These columns describe when the COVID-19 cases and deaths were reported. They are used for yearly, monthly, and quarterly analysis.
COVID-19 indicators	- cases - deaths	These columns contain the daily number of reported COVID-19 cases and deaths.
Country information	- countriesAndTerritories - geold - countryterritoryCode - continentExp	These columns identify the reporting country and are used for grouping, comparison, and mapping.
Population information	- popData2020	This column shows the country population in 2020. It is used to compare countries more fairly by calculating values relative to population size.

The dataset includes variables such as country name, reporting date, number of new cases, number of deaths, population size, and geographical information.

2.1.1 Data Preprocessing

The dataset was preprocessed to ensure that the data was suitable for comparison, aggregation, and visualization. The preprocessing steps focused on preparing the time-related columns, handling missing or inconsistent values, and then transforming the data into a format that supports the research questions.

Table 2 Preprocessing Summary for Daily Cases and Deaths Dataset

Finding	Step	Method	Motivation
Negative values: - cases contains 14 negative values - deaths contains 17 negative values	COVID-19 indicators	Convert negative values to NaN	Negative values represent retroactive historical corrections (adjusting for previous overcounting), leaving them as large negative numbers creates unnatural downward drops on charts. Converting them to NaN allows you to treat them exactly like missing records.
Missing values: - cases contains 93 (0.3%) missing values - deaths contains 292 (1%) missing values	Missing value handling	1. Convert dates and sort chronologically 2. Handle Edge Cases with Zero-Filling 3. Country-Level Grouping 4. Apply Grouped Linear Interpolation	1. The dataset is currently organized in reverse-chronological order (newest dates first). Time-series interpolation only works correctly if the data flows forward from oldest to newest. 2. If a missing or negative value occurs on the very first recorded day for a country, linear interpolation cannot look backward to find a starting point. Adding zero values catches these rare edge cases. 3. Deleting rows containing missing or negative data would rupture the continuous chronological sequence required for tracking an epidemic over time. 4. Interpolation must be localized within each country entity via a group by operation to prevent regional epidemiological data from bleeding across unrelated nations. 5. Since pandemic data moves in waves and sequential trends, linear interpolation provides an empirically grounded estimate of missing daily counts based on surrounding reporting days, ensuring smooth curves for analytical tracking.

Why are there negative values?

Overcounting Corrections: Public health agencies often review their historical records. If a country realizes it accidentally double-counted cases or misclassified non-COVID deaths as COVID deaths during previous weeks, they issue a correction. Instead of retroactively changing every single past day in their database, they simply apply a negative number to the current day to adjust the running cumulative total. For example, looking into the dataset, France reported -348,846 cases on May 20, 2021. This was a massive historical data correction made by the French government to clean up duplicated patient files from earlier months.

2.2 Vaccination Dataset

This dataset contains 681,330 rows and 18 columns vaccination information across EU/EEA countries.

Table 3 Column Categories in the Vaccination Dataset

Column Category	Column Name	Description
Time Information	- YearWeekISO	This column shows the reporting year and week. It is used to analyze vaccination progress over time.
Country and region information	- ReportingCountry - Region	These columns identify the reporting country and region. They are used for country-level vaccination comparisons.
Population information	- FirstDose - SecondDose - DoseAdditional1 - DoseAdditional2 - DoseAdditional3 - DoseAdditional4 - DoseAdditional5 - UnknownDose	These columns describe the number of people who received different vaccine doses, including first dose, second dose, additional/ booster doses, and unknown doses.
Vaccine refusal information	- FirstDoseRefused	This column shows the number of people who refused the first dose. It is used to investigate vaccine skepticism.
Target group information	- TargetGroup	This column describes the population group receiving the vaccine, such as age groups or priority groups.
Vaccine brand information	- Vaccine	This column identifies the vaccine brand used in each record. It is used to analyze which vaccine brands were most commonly used across EU/EEA countries.
Vaccine supply information	- NumberDosesReceived - NumberDosesExported	These columns describe the number of vaccine doses received and exported by each country. They provide additional context about vaccine availability.

2.2.1 Data Preprocessing

We found missing values in some columns in the dataset.

What is Denominator?

The Denominator represents the size of the specific target group within that country (e.g., the number of healthcare workers or the population of the Age70_79).

Table 4 Preprocessing Summary for Vaccination Dataset

Finding	Step	Method	Motivation
Missing values: - Denominator contains 319,779 (46.9%) missing values - NumberDosesReceived contains 595,499 (87.4%) missing values - NumberDosesExported contains 590,036 (86.8%) missing values - FirstDoseRefused contains 679,883 (99.78%) missing values	Missing value handling	1. Impute missing values with 0 for NumberDosesReceived, NumberDosesExported, and FirstDoseRefused column 2. Keep Denominator as NaN	1. Deleting rows containing missing or negative data would rupture the continuous chronological sequence required for tracking an epidemic over time. 2. In public vaccine logistics reporting, if a row tracking a specific vaccine brand (e.g., Novavax) for a specific age group during a specific week is blank under doses received or exported, it is because no doses were received or exported. 3. Missing values were interpreted as no reported first-dose refusal cases, which supports the analysis of vaccine skepticism across countries. 4. Countries often lacked accurate, dynamic data for specific occupational target groups by age or category like healthcare workers. Filling this column with a static value (like 0) would result in critical "Division by Zero" (inf) errors when computing vaccine uptake rates () later. Imputing with a mean or median would create false population statistics across unrelated countries. Leaving these entries as NaN preserves data authenticity, allowing us to drop or circumvent them dynamically and safely.

2.3 Hospital and ICU Dataset

This dataset contains 25,100 rows and 7 columns information about hospitalization and ICU occupancy related to COVID-19.

Table 5 Column Categories in the Hospital and ICU Admission Rates Dataset

Column Category	Column Name	Description
Time Information	- year_week - date	These columns describe the reporting period and reporting date. They are used to analyze hospitalization and ICU trends over time.
Country information	- country	This column identifies the reporting country and is used for country-level comparisons across the EU/EEA region.
Healthcare indicator information	- indicator	This column specifies the type of healthcare measurement, such as hospital admissions, ICU admissions, hospital occupancy, or ICU occupancy.
Measurement values	- value	This column contains the reported numerical values for each healthcare indicator. It is used to measure the impact of COVID-19 on healthcare systems.
Data source information	- source, url	These columns provide information about the origin of the reported healthcare data. They improve data traceability and transparency but are not directly used in the main analysis.

2.3.1 Data Preprocessing

We found 100% missing values in a column in the dataset.

Table 6 Preprocessing Summary for Hospital and ICU Dataset

Finding	Step	Method	Motivation
Missing values: - url contain 25,100 (100%) missing values	Missing value handling	Remove column url	Because an entirely blank column provides no statistical variance or administrative value, and since external data reference links are completely irrelevant to resolving the core assignment questions so it is better to remove that column.

2.4 Data Transformation

After the preprocessing stage, several data transformation steps were applied to prepare the datasets for analysis, visualization, and statistical comparison that supports the research questions.

The main steps included sorting the data chronologically, creating time-based variables, calculating population-normalized rates, applying log transformations, and generating aggregated summary tables.

1. First, the datasets were sorted by country, date, and relevant indicators to ensure that the observations followed a logical chronological order. Then, time-related variables such as year, week, quarter, and year-quarter labels were created to support yearly, weekly, and quarterly analysis. These variables made it easier to compare COVID-19 trends across different time periods.
2. Next, population-normalized variables were created, such as cases per 100,000 people, deaths per 100,000 people, first-dose vaccination rate, second-dose vaccination rate, and first-dose refusal rate. These transformations were important because EU/EEA countries have different population sizes, and raw numbers alone may lead to unfair comparisons.
3. Several summary tables were created by grouping the data by country, time period, vaccine brand, target group, or healthcare indicator. These summary tables helped simplify the original daily or weekly datasets into structured forms that were easier to use for plots, tables, and statistical comparison.

3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to better understand the dataset structure and pattern before conducting the main analyses. The EDA process focused on identifying trends, distributions, geographical differences, and potential relationships between COVID-19 cases, deaths, vaccination coverage, and healthcare indicators across EU/EEA countries.

Time-series plots were used to observe changes in COVID-19 cases, deaths, vaccination progress, and hospital occupancy over time. Bar charts and grouped comparisons were used to identify countries with the highest and lowest values in different categories.

The exploratory analysis also included examining vaccine usage by brand and target group, identifying countries with higher first-dose refusal rates, and analyzing healthcare indicators such as hospital occupancy and ICU admissions. In addition, aggregated quarterly and yearly summary tables were used to simplify large daily and weekly datasets into more interpretable forms for visualization and comparison.

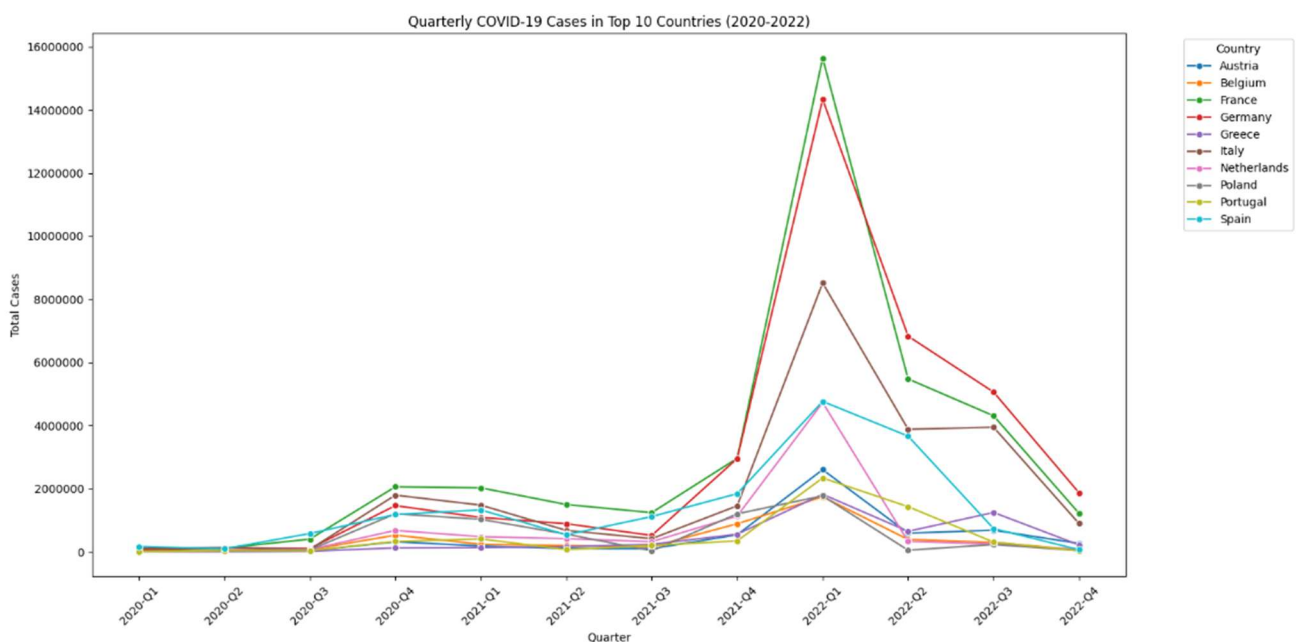
Overall, the EDA stage provided important insights into the datasets and helped guide the selection of appropriate methods, visualizations, and statistical analyses for answering the research questions in the following sections.

4 Cases and Deaths Data: COVID-19 Cases and Death Analysis

The following subsections present the analyses, visualizations, and interpretations related to Question 1 and Question 2.

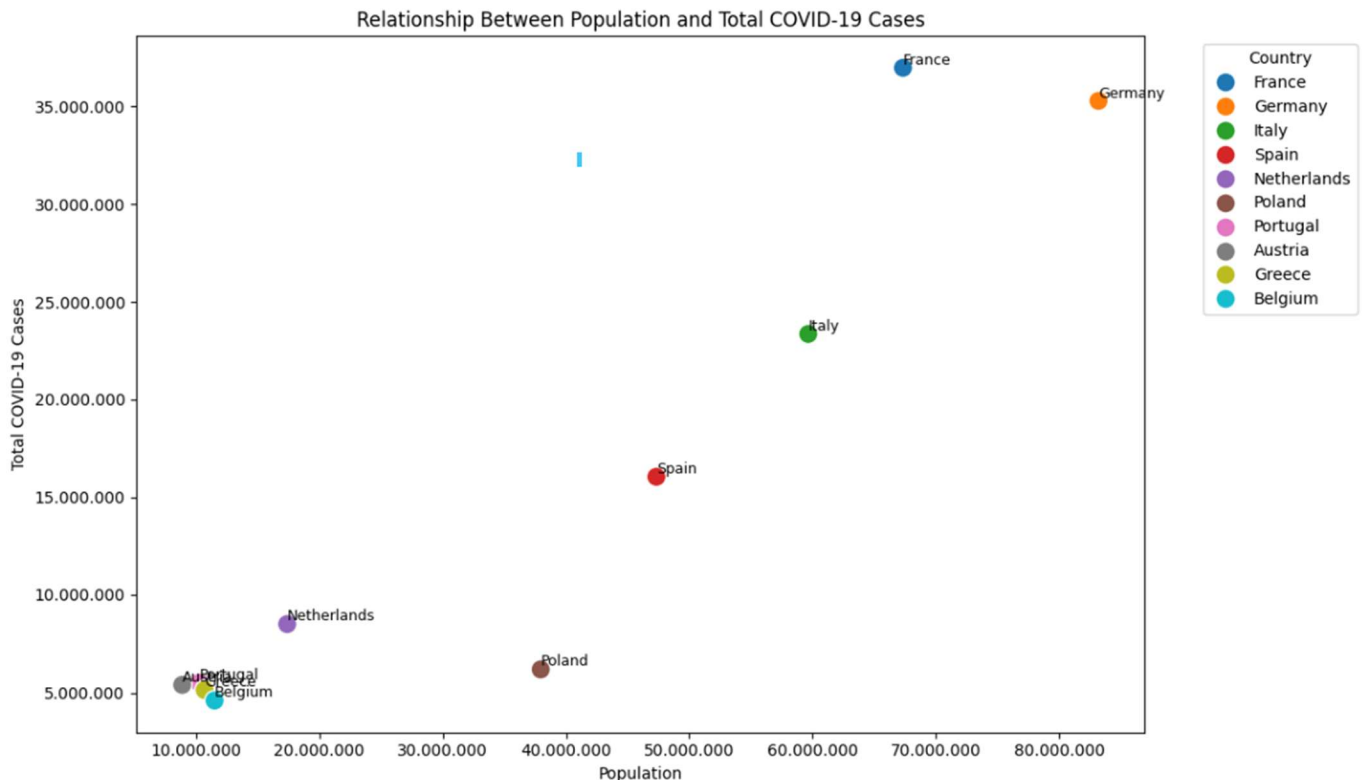
4.1 Q1.1: Top countries with the highest cases by quarter.

year_quarter	2020-Q1	2020-Q2	2020-Q3	2020-Q4	2021-Q1	2021-Q2	2021-Q3	2021-Q4	2022-Q1	2022-Q2	2022-Q3	2022-Q4	Total Cases
Country													
France	52.133	113.784	398.734	2.056.890	2.023.998	1.495.649	1.238.131	2.959.368	15.642.043	5,474,911	4,307,356	1,215,517	36,978,514
Germany	74.061	120.893	98.995	1,460,483	1,085,469	888,207	514,274	2,952,749	14,337,869	6.830.657	5.058.149	1.865.884	35,287,690
Italy	105.792	135.232	74.283	1,792,305	1,476,278	676,465	412,446	1,453,328	8,516,671	3,880,757	3,944,167	892,402	23,360,126
Spain	167.645	93.346	575.532	1,180,775	1,325,289	535,698	1,114,164	1,838,335	4,762,082	3,662,976	731,828	65,031	16,052,700
Netherlands	12.449	37.598	70.530	674,961	476,152	412,910	318,118	1,128,856	4,727,250	325,309	246,712	63,860	8,494,705
Poland	2.311	32.082	57.121	1,203,364	1,026,839	558,195	27,159	1,201,144	1,770,193	47,187	229,187	34,780	6,189,562
Portugal	7.251	35,673	34,439	323,243	406,514	66,331	198,724	342,251	2,336,007	1,427,465	304,946	31,638	5,514,482
Austria	7.312	7.253	28,065	314,780	182,007	101,655	95,232	535,893	2,599,732	586,570	682,264	261,399	5,402,162
Greece	1.314	2,095	15,066	120,375	130,716	158,767	233,311	555,086	1,819,576	646,073	1,243,690	215,008	5,141,078
Belgium	16.977	45,007	63,834	524,222	236,586	193,903	159,581	885,792	1,740,620	388,325	296,782	55,667	4,607,296



4.2 Q1.2: The relation between the high number of cases and the country's population.

Correlation between population and total cases:
0.932



Observation:

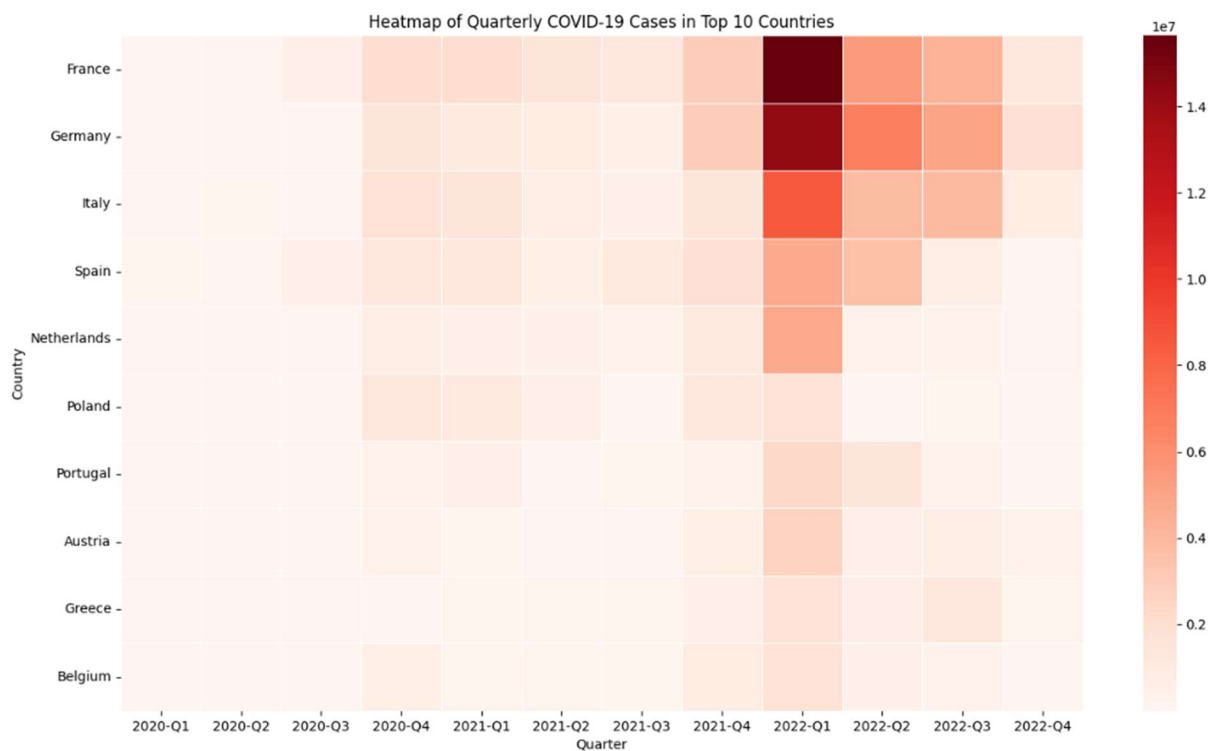
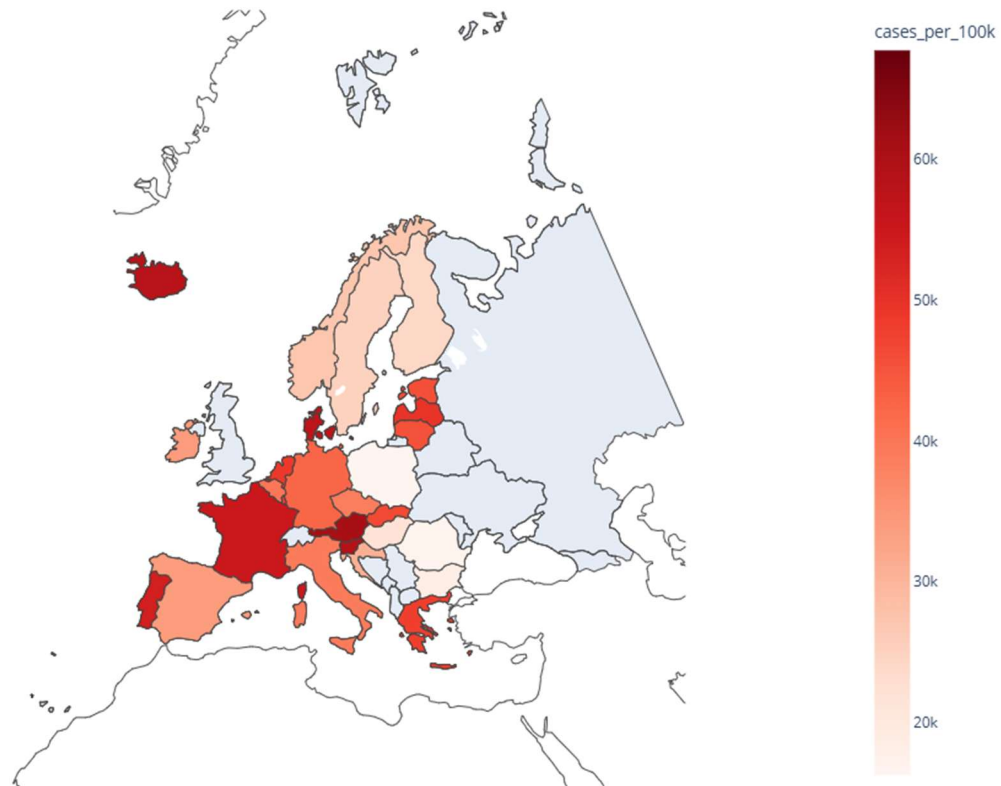
- The scatter plot shows a clear upward trend between population size and total COVID-19 cases, indicating that countries with larger populations generally reported higher numbers of cases. Most data points follow a positive linear pattern, although the relationship is not perfectly linear because some smaller countries also experienced relatively high case numbers.
- The Pearson correlation coefficient between population and total COVID-19 cases is **0.92**, which indicates a **very strong positive correlation**. This means that population size is strongly associated with the number of reported COVID-19 cases. However, since the correlation is not exactly 1.0, other factors such as transmission intensity, population density, mobility, and public health measures also likely influenced the spread of COVID-19.

Conclusion:

The analysis suggests that countries with larger populations tended to report higher numbers of COVID-19 cases, although population size alone does not fully explain the variation in case numbers across countries.

4.3 Q1.3: The connection between countries regarding their geographical locations and the period.

COVID-19 Cases per 100k Population by Country (2020-2022)



Observation:

- The geographical visualization shows that most countries with the highest COVID-19 case numbers are located in Western and Central Europe. These countries are geographically close

and highly connected through trade, tourism, and open-border travel within the European Union and the Schengen Area. This regional proximity and mobility may have contributed to similar transmission patterns across neighboring countries. When adjusted for population size, some smaller countries also showed high infection rates, indicating that raw case counts alone do not fully reflect the relative pandemic impact.

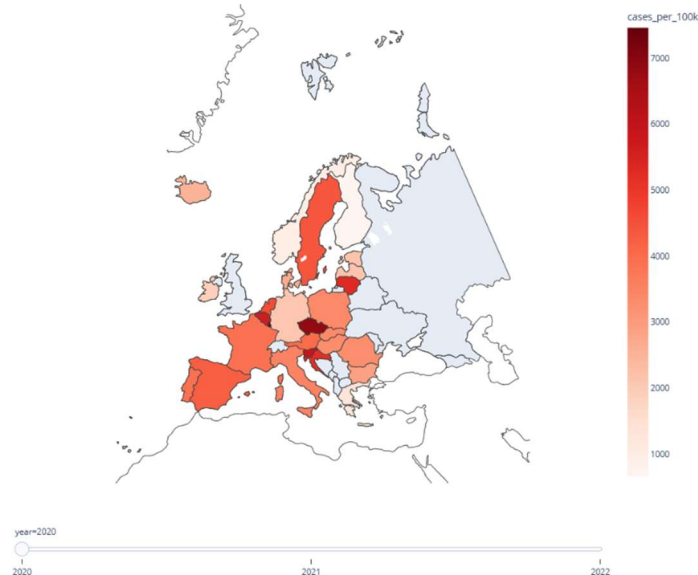
- The quarterly heatmap shows that many of these countries experienced peak case numbers during similar periods, especially in Q4 2021 and Q1 2022, which corresponds to the Omicron wave. This suggests synchronized regional outbreaks across geographically connected countries.

Conclusion:

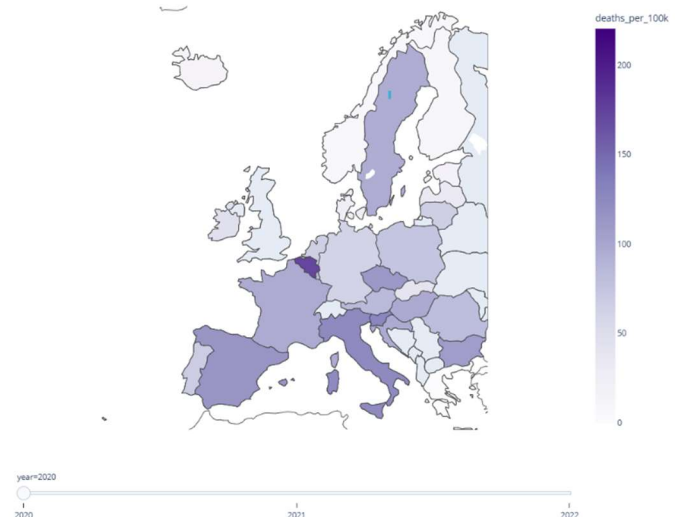
The analysis shows both geographical and temporal synchronization among the highly affected European countries. This indicates that regional proximity, mobility, and shared transmission dynamics played important roles in the spread of COVID-19 across Europe.

4.4 Q2: Map visualization of cases and deaths (2020-2022).

COVID-19 Cases per 100k Population in Europe (2020-2022)



COVID-19 Deaths per 100k Population in Europe (2020-2022)



Observation:

- The darker the color on the map, the higher the proportion of COVID-19 cases or deaths relative to the country's population. Therefore, the map not only shows raw total cases or deaths, but also reflects the relative pandemic burden after adjusting for population size. As a result, smaller countries can still appear very dark if they experienced high infection or mortality rates relative to their populations
- The COVID-19 cases and deaths were concentrated mainly in Western and Central Europe throughout 2020–2022
- Countries such as France, Germany, Italy, and Spain consistently recorded high case numbers, which may be influenced by their large populations, high mobility, and strong international connectivity
- The geographical spread of cases became more widespread over time, particularly in 2021 and 2022. Many countries showed substantial increases in case intensity during 2022, likely associated with the rapid spread of the Omicron variant across Europe
- The deaths map indicates that several countries experienced relatively high mortality rates during 2020 and 2021. However, despite high case numbers in 2022, many countries showed comparatively lower death intensity relative to earlier periods. This may reflect the effects of vaccination programs, improved healthcare responses, and increased population immunity
- Neighboring countries often displayed similar pandemic patterns, suggesting regional transmission dynamics across Europe

Conclusion:

The visualizations indicate that geographical proximity, cross-border mobility, and synchronized pandemic waves played important roles in the spread and impact of COVID-19 across European countries.

5 Vaccination Data: Vaccination Brand and Distribution Analysis

5.1 Q3: The top 3 popular vaccine brands used across the EU/EEA and at the country level.

	Vaccine	total_first_dose	total_second_dose
0	COM	455,883,035	433,530,979
1	UNK	289,412,190	279,408,790
2	AZ	72,640,842	64,506,477

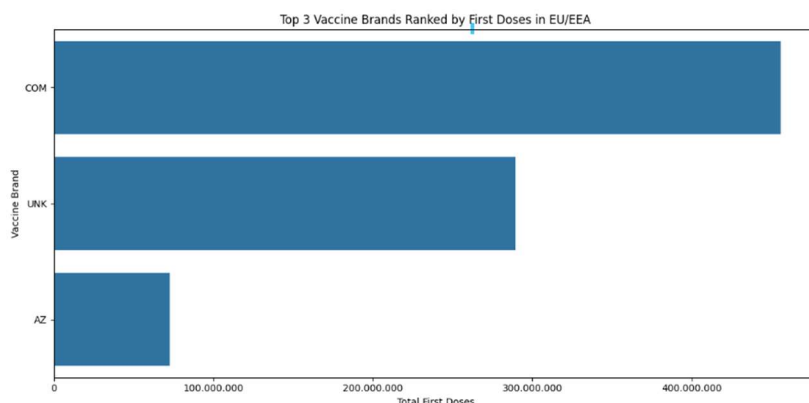
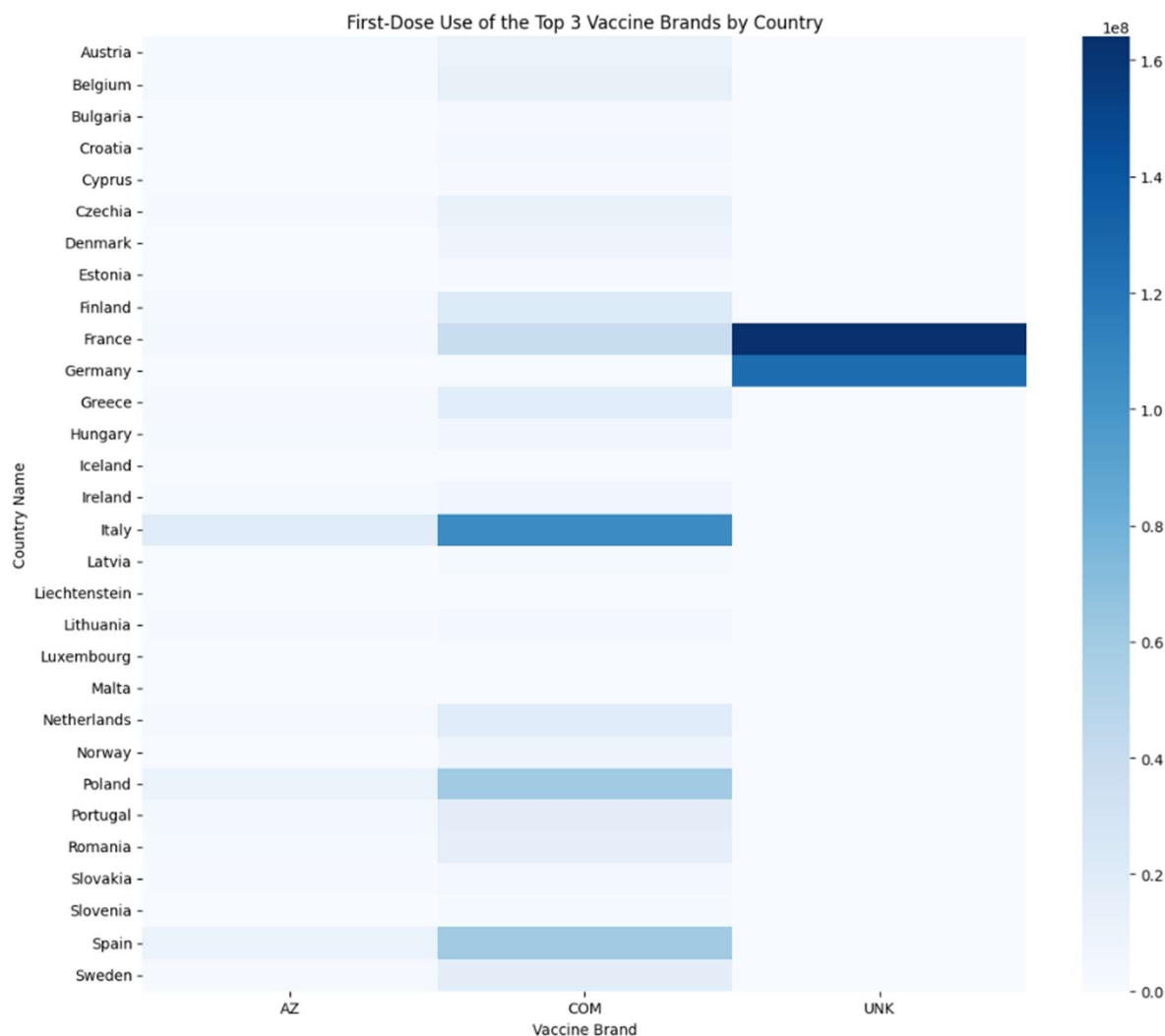


Table 7 Vaccine Brand

Code	Vaccine Brand
COM	Comirnaty/ Pfizer-BioNTech
MOD	Moderna/ Spikevax
AZ	AstraZeneca/ Vaxzevria
JANSS	Janssen
NVXD	Novavax
UNK	Unknown vaccine brand



Observation:

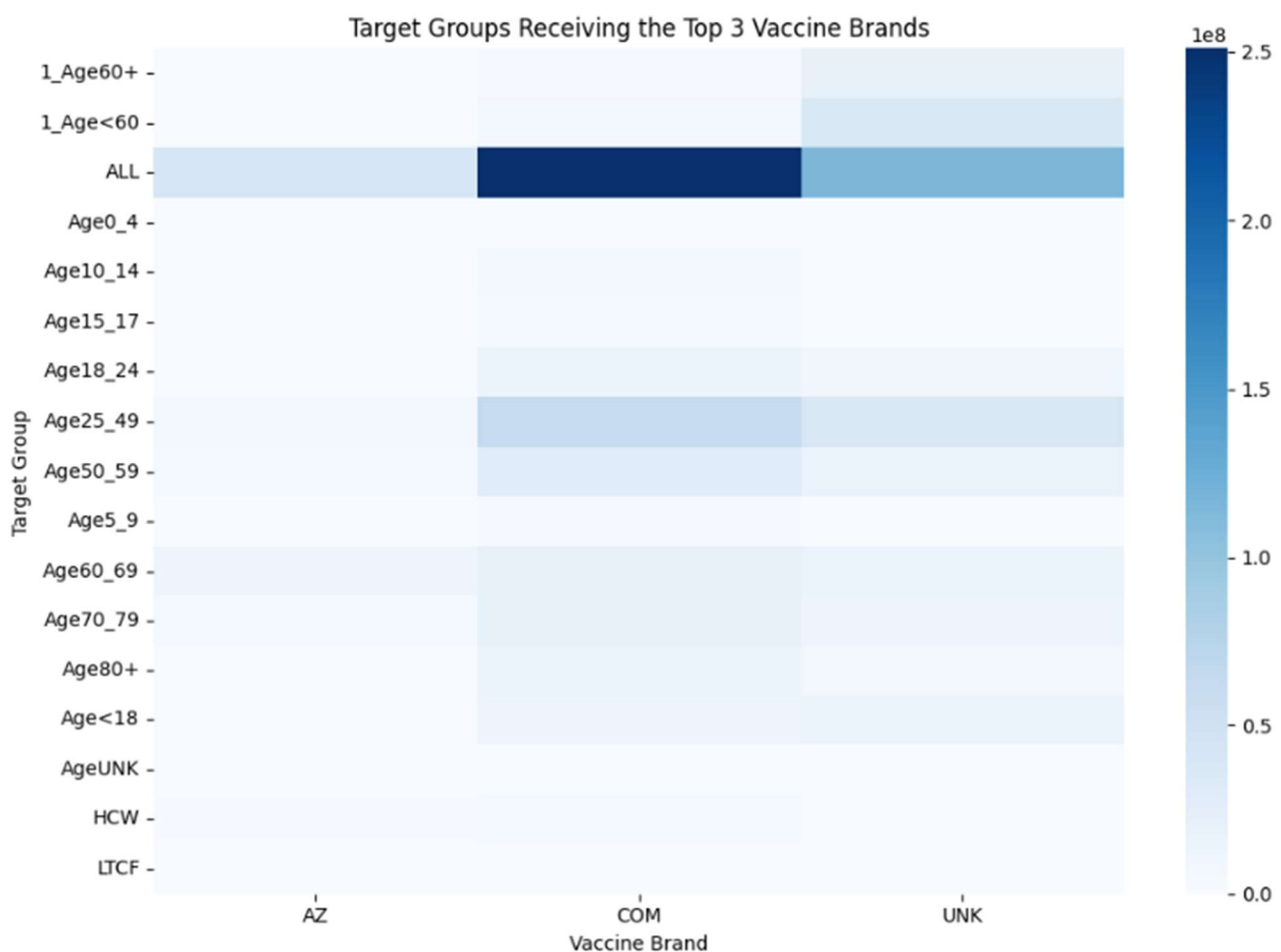
- The analysis shows that the three most popular vaccine brands by total number of first doses used across the EU/EEA were Comirnaty / Pfizer-BioNTech (COM), Unknown vaccine brand (UNK), AstraZeneca / Vaxzevria (AZ).
- The results show that several vaccine brands were consistently popular across many EU/EEA countries, indicating widespread distribution and adoption during the vaccination campaign.
- The country-level heatmap suggests that the overall EU/EEA popularity trend was generally reflected in many countries, especially for the vaccine brand with the highest first-dose count.
- Exception: some countries showed different vaccination patterns, where certain vaccine brands were used more or less extensively compared to the overall EU/EEA trend.
- These differences indicate that vaccine popularity was influenced not only by overall EU/EEA distribution but also by country-level factors such as vaccine availability, national vaccination policies, healthcare strategies, and public acceptance.

- The presence of the UNK category indicates that some vaccination records did not specify the vaccine brand, so this category should be interpreted carefully when comparing vaccine popularity across countries.

Conclusion:

The analysis indicates that Pfizer/BioNTech was the dominant vaccine brand across the EU/EEA and also remained the most widely used vaccine at the country level. However, some exceptions remain, as a few countries recorded UNK (Unknown vaccine brand) as the most frequently recorded category. This suggests that certain vaccination records did not specify the vaccine brand, which may affect the completeness and consistency of the country-level comparison.

5.2 Q4: Target groups received the popular vaccine brands in each country



Observation:

- The analysis focuses on first-dose vaccination because first doses better reflect vaccination prioritization and allocation strategies across target groups during the initial rollout phases
- The heatmap shows that the dominant target group for most countries and vaccine brands was ALL, indicating that the top 3 vaccine brands were distributed differently across target groups.
- Darker color intensity indicates target groups that received higher numbers of first doses for a particular vaccine brand.

- Certain vaccine brands appear to be more strongly associated with specific target groups, indicating differences in vaccine allocation and usage policies.

Conclusion:

The visualization highlights clear patterns in how the most popular vaccine brands were administered across different target populations in the EU/EEA.

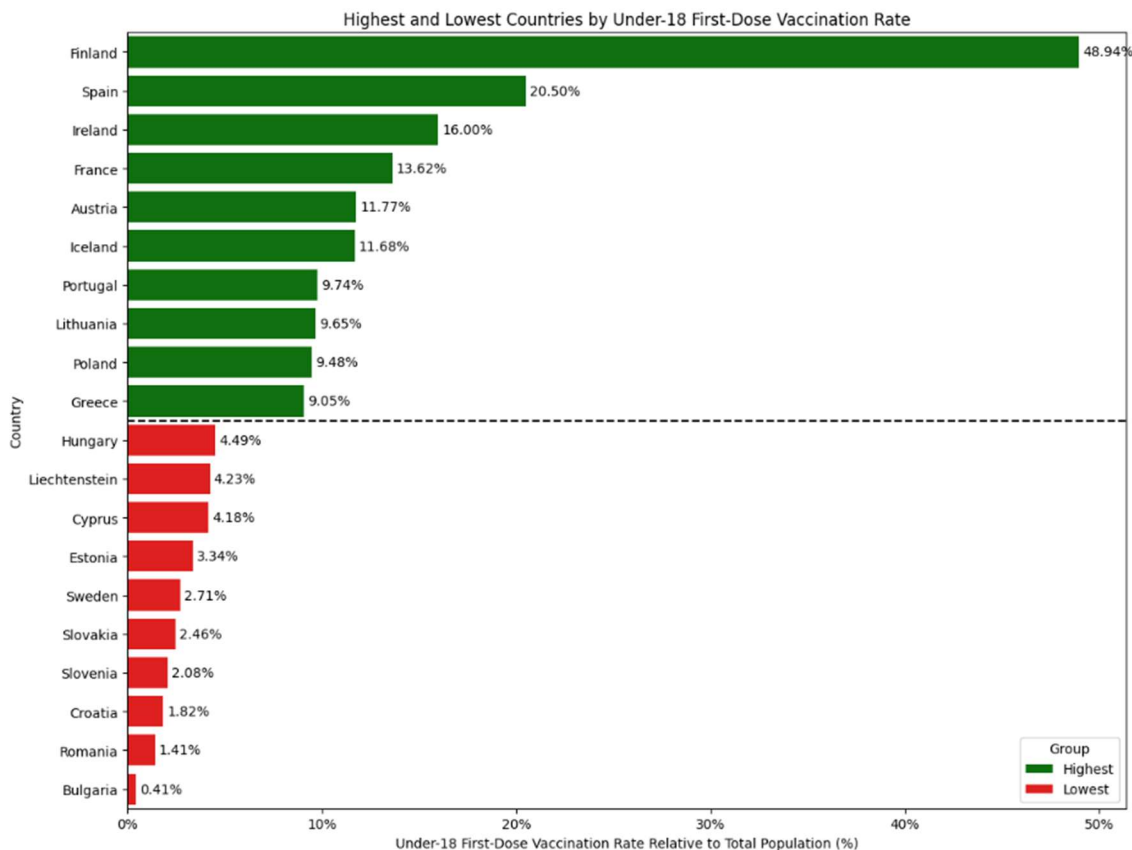
6 Vaccination Data: Vaccine Skepticism and Hospitalization

6.1 Q5: The most skeptical countries towards the first dose of the COVID-19 vaccine.

Only one country reported first-dose refusal value, **Iceland**, making the dataset insufficient for a reliable statistical comparison or correlation analysis. Therefore, it is difficult to conclude that first-dose vaccine skepticism had a significant impact on hospitalization levels. However, when considering the broader patterns across the cases, vaccination, and hospitalization datasets, vaccination coverage may still have influenced hospitalization outcomes. The available refusal data does not provide enough evidence to determine whether first-dose vaccine skepticism directly affected hospitalization levels across EU/EEA countries.

7 Vaccination Data: Vaccination Coverage and Demographic Analysis

7.1 Q6: Countries with the most and the least vaccinated people under 18 regarding their total populations.



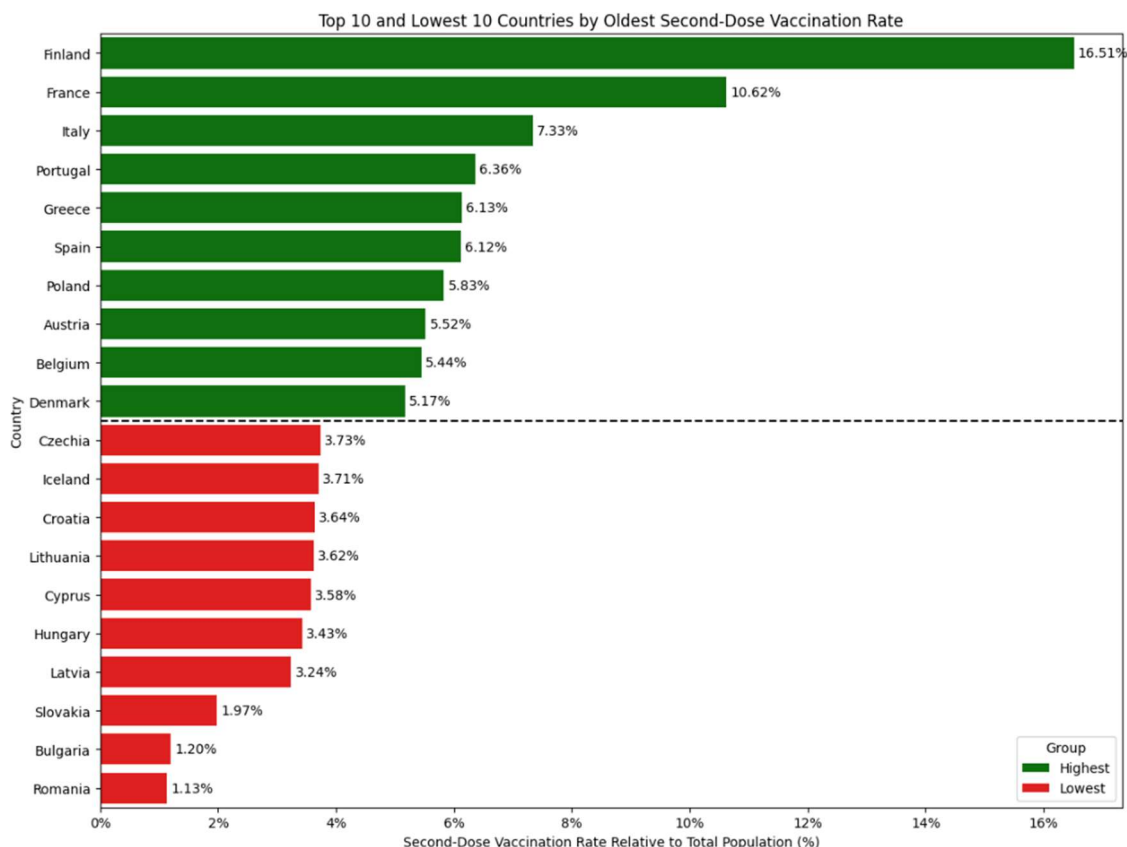
Observations:

- The analysis shows clear differences in under-18 first-dose vaccination rates across EU/EEA countries.
- The analysis used first-dose vaccination counts to estimate the number of vaccinated individuals under 18 years old. First-dose data were selected because combining first and second doses could lead to double counting of vaccinated individuals and overestimate vaccination coverage.
- Countries with the highest rates had a larger proportion of children and adolescents receiving at least one vaccine dose relative to the total national population.
- Countries with the lowest rates had much smaller under-18 first-dose coverage, suggesting limited rollout, lower uptake, or different national policies for younger age groups.
- The color gradient further highlights the contrast between highly vaccinated and less vaccinated countries, while the separation line clearly distinguishes the highest and lowest groups.

Conclusion:

The results indicate that under-18 vaccination was not uniform across the EU/EEA and was likely influenced by policy decisions, vaccine availability, and public acceptance.

7.2 Q7: The countries with the oldest vaccinated population for the second dose vaccine in relation to their total population.



Observation:

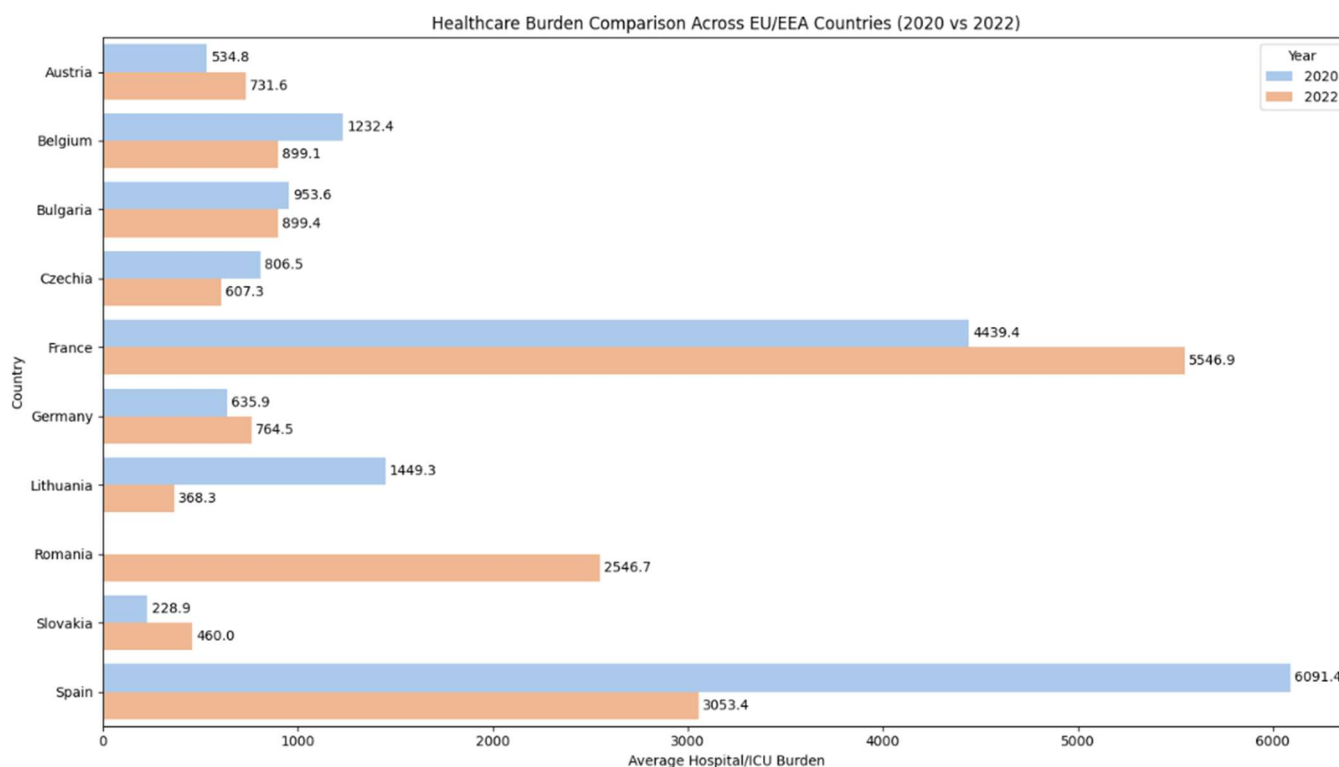
- The analysis shows clear differences in second-dose vaccination coverage among the oldest population group (Age80+) across EU/EEA countries
- Countries in the highest-ranked group achieved substantially higher vaccination rates among individuals aged 80 to their total populations. This suggests that these countries were more successful in prioritizing and fully vaccinating their oldest and most vulnerable populations during the COVID-19 vaccination campaigns.
- In contrast, the lowest-ranked countries displayed considerably lower second-dose vaccination rates for the Age80+ group. These differences may reflect variations in healthcare accessibility, vaccination rollout efficiency, vaccine acceptance, demographic structure, or national vaccination policies.
- The visualization also highlights the contrast between countries with strong elderly vaccination coverage and those with relatively limited coverage. Since elderly populations were among the highest-risk groups for severe COVID-19 outcomes, higher second-dose coverage among individuals aged 80 and above may indicate stronger protection for vulnerable populations.

Conclusion:

The results suggest that EU/EEA countries differed substantially in their success at fully vaccinating the oldest population groups during the pandemic.

8 Hospital and ICU Data: Healthcare System Impact

8.1 Q8: The most affected countries' health care by the coronavirus pandemic in 2020.



Observation:

- The plot shows the top 10 countries with the highest average healthcare burden across 2020 and 2022 combined.
- The average healthcare burden was calculated by taking the mean value of hospital/ ICU values for each country within the years 2020 and 2022.
- Romania has no record of hospital/ ICU for 2020.
- Higher average values indicate greater pressure on healthcare systems during the pandemic.
- In 2022, healthcare burden statistics generally became lower or more stable compared to 2020 in several EU/EEA countries. This suggests that healthcare systems were better able to manage the pandemic after the initial outbreak period.
- The reduced burden in 2022 may be associated with
 - widespread vaccination coverage
 - improved clinical treatment
 - greater healthcare preparedness
 - higher population immunity
- Some countries continued to experience a relatively high healthcare burden in 2022, indicating that COVID-19 still placed pressure on healthcare systems despite vaccination campaigns and public health interventions

Conclusion:

The comparison suggests that while the pandemic continued to affect healthcare systems in 2022, the impact was generally less severe and more manageable than during the early pandemic period in 2020.

9 Additional Questions

This further analysis mainly focused on 2022 because vaccination programs and booster campaigns were more fully implemented during this period. Using a consistent time frame across datasets also allows more reliable comparison between vaccination coverage, COVID-19 outcomes, and healthcare burden indicators.

For the statistical correlation analysis, the Pearson correlation method was used to measure the strength and direction of the linear relationship between two numerical variables.

For plotting, the Scatter plot + regression line was the best option because it is the most appropriate for correlation analysis.

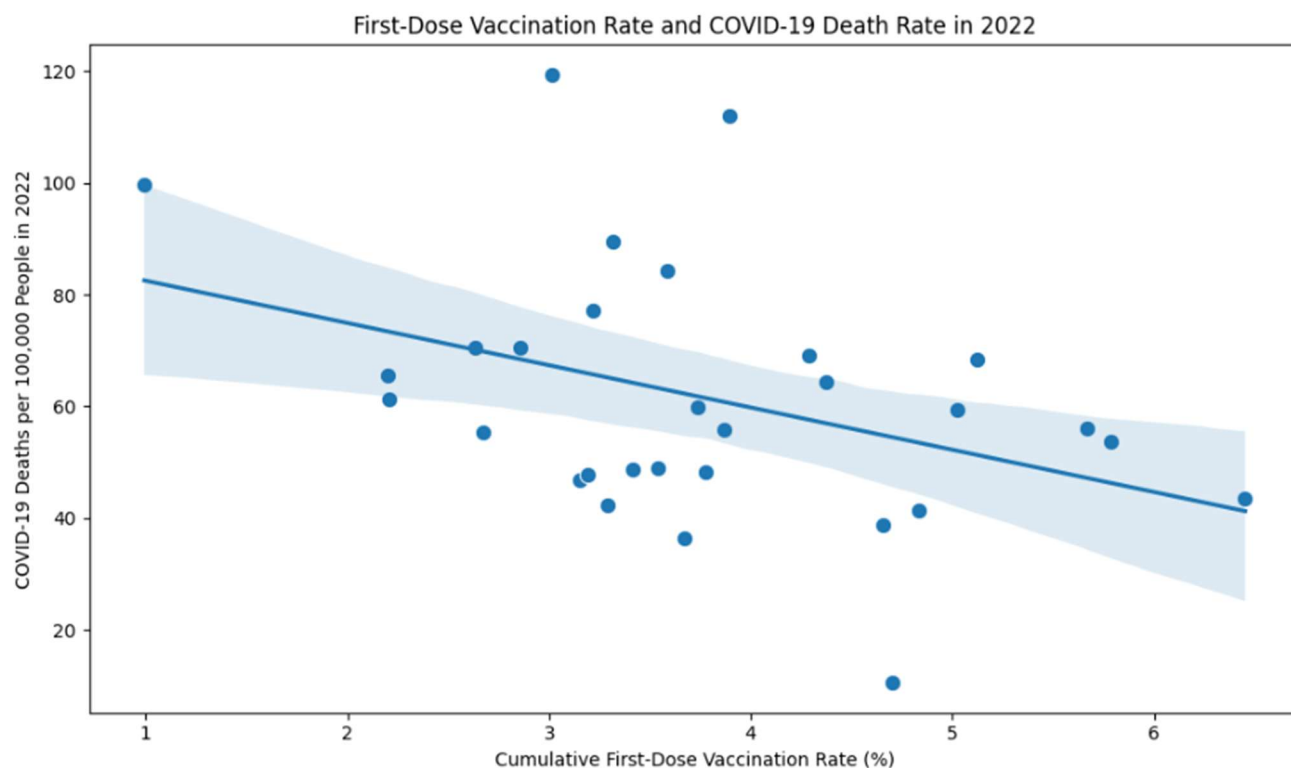
Table 8 Additional Research Question

No	Research Question	Datasets Used	Method	Suggested Plot	Motivation and Public Value
1	Did country with higher first-dose vaccination rates experience lower COVID-19 death rates?	Vaccination dataset +cases and deaths dataset	- Merge country-level vaccination rates with country-level death rates	Scatter plot with regression line	- This question is worth investigating because it helps the public understand whether higher vaccination coverage was

No	Research Question	Datasets Used	Method	Suggested Plot	Motivation and Public Value
			- Use correlation analysis		- associated with lower COVID-19 mortality. - It provides useful insight into the potential public health value of vaccination campaigns across EU/EEA countries.
2	Did countries with higher booster or additional-dose uptake show lower hospital or ICU pressure in later pandemic periods?	Vaccination dataset + hospital and ICU dataset	- Aggregate additional-dose vaccination rates by country and compare them with hospital/ICU indicators in 2022. - Use correlation analysis	Scatter plot with regression line	- This question is important because booster vaccination was introduced to reduce severe disease and pressure on healthcare systems. - The results may provide public knowledge about whether higher booster uptake was associated with reduced healthcare burden.
3	Did countries with high COVID-19 case rates also experience high healthcare burden?	Cases and deaths dataset + hospital and ICU dataset	- Aggregate country-level COVID-19 case rates and compare them with hospital and ICU occupancy indicators using correlation analysis.	Scatter plot or heatmap	- This question is valuable because it investigates whether countries with larger outbreaks also experienced greater pressure on healthcare systems. - The findings may help the public understand how increases in infection rates can affect hospital capacity and healthcare resources during a pandemic.

9.1 Additional Questions 1: Did country with higher first-dose vaccination rates experience lower COVID-19 death rates?

Correlation between first-dose vaccination rate and death rate: -0.389



Observation:

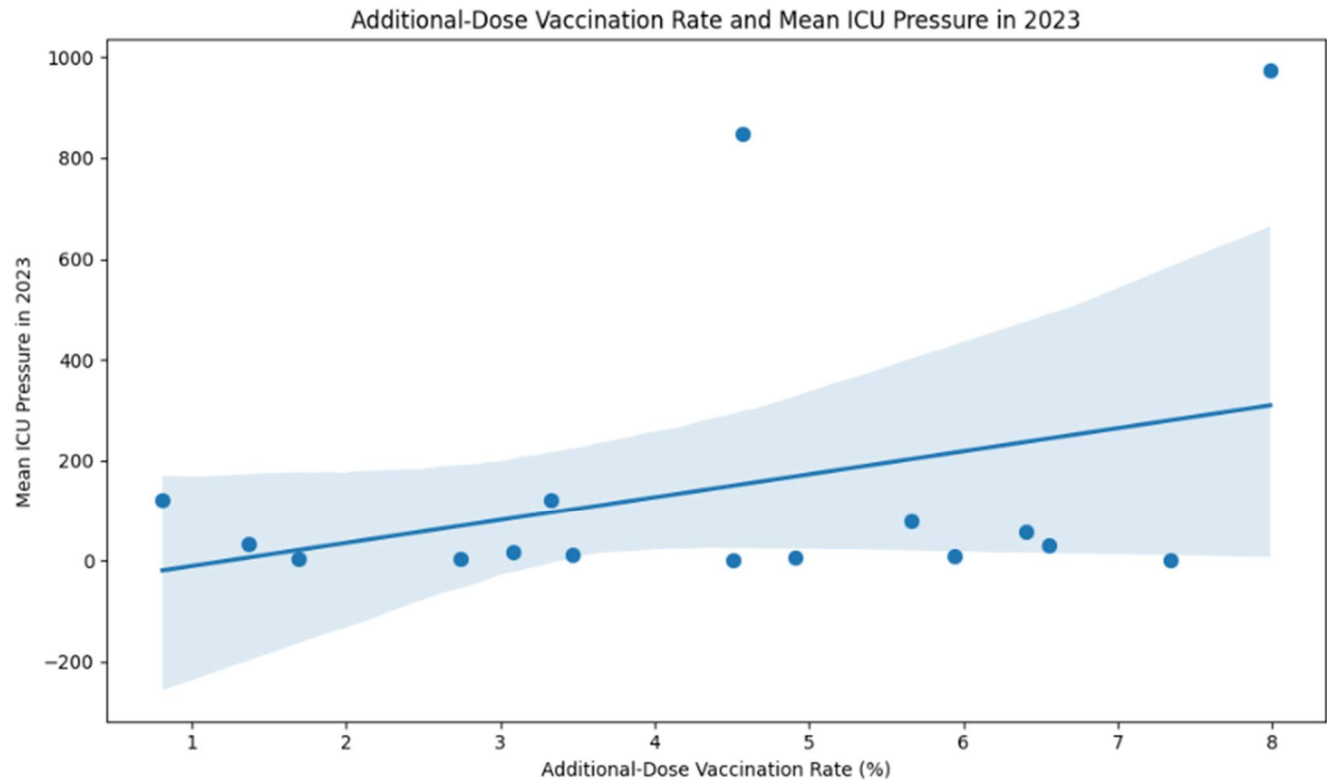
- The scatter plot shows a downward trend between cumulative first-dose vaccination rate and COVID-19 death rate in 2022.
- Each dot represents one EU/EEA country. Countries with higher first-dose vaccination coverage generally tended to have lower death rates per 100,000 people.
- The Pearson correlation value is -0.389, which indicates a weak-to-moderate negative relationship. The relationship is not very strong because the data points are still widely spread.
- This means vaccination coverage alone does not fully explain differences in death rates. Other factors such as age structure, healthcare quality, public health policies, virus variants, and reporting differences may also affect death rates.

Conclusion:

The analysis suggests that higher first-dose vaccination coverage was associated with lower COVID-19 death rates in 2022. However, the relationship was only weak to moderate, so the result should be interpreted as an association, not direct causation.

9.2 Additional Questions 2: Did countries with higher booster or additional-dose uptake show lower hospital or ICU pressure in later pandemic periods?

Correlation between additional-dose rate and mean ICU pressure: 0.328



Observation:

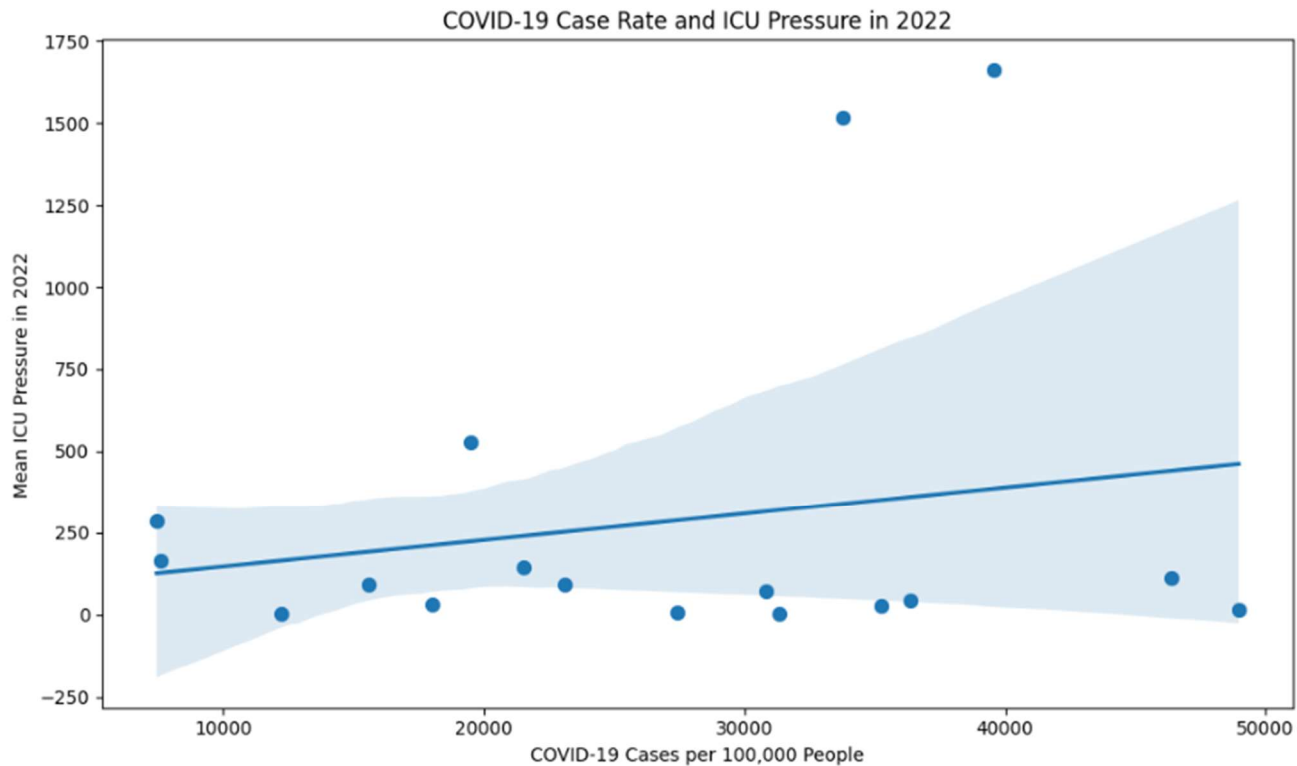
- The scatter plot shows a weak relationship between additional-dose vaccination rates and ICU pressure in 2023.
- Countries with higher additional-dose uptake did not always experience lower ICU pressure.
- Some countries with high additional-dose vaccination rates still showed relatively high ICU pressure.
- The data points are widely spread, indicating large variation between countries.
- The regression line shows only a slight trend, meaning that additional-dose uptake alone was not strongly associated with ICU pressure levels.
- The wide confidence interval suggests uncertainty in the relationship between the two variables.
- These results indicate that ICU pressure in 2023 was likely influenced by multiple factors besides vaccination, such as healthcare system capacity, age distribution, virus variants, and public health measures.

Conclusion:

The analysis did not show a strong relationship between additional-dose vaccination uptake and ICU pressure in 2023. Although additional doses may have helped reduce severe COVID-19 outcomes in some countries, ICU burden across EU/EEA countries was still affected by many other factors beyond vaccination coverage alone.

9.3 Additional Questions 3: Did countries with high COVID-19 case rates also experience high healthcare burden?

Correlation between case rate and healthcare burden: 0.200



Observation:

- The scatter plot shows a slight upward trend between COVID-19 case rates and ICU pressure in 2022.
- The Pearson correlation value is 0.200, which indicates a weak positive relationship between the two variables.
- Countries with higher COVID-19 case rates sometimes experienced higher ICU pressure, but the relationship was not strong.
- Several countries with very high case rates still showed relatively low ICU pressure, indicating that high infection rates did not always lead to severe healthcare burden.
- A few countries experienced extremely high ICU pressure compared with others, which may have influenced the regression line and overall correlation.
- The data points are widely spread across the plot, suggesting large variation between countries.
- The wide confidence interval indicates uncertainty in the relationship between case rates and ICU pressure.
- These results suggest that ICU burden in 2022 was influenced not only by infection rates, but also by other factors such as vaccination coverage, healthcare system capacity, population age structure, virus severity, and public health policies.

Conclusion:

Overall, the analysis found only a weak positive relationship between COVID-19 case rates and ICU pressure in 2022. Although countries with higher case rates sometimes experienced greater healthcare burden, infection rates alone did not strongly explain ICU pressure differences across EU/EEA countries.

10 Discussion

- The results show that COVID-19 outcomes across EU/EEA countries were shaped by several connected factors rather than by one single cause. Population size was strongly related to the total number of reported cases, but it did not fully explain the differences between countries. Some smaller countries also showed high case rates when the data was adjusted by population size, indicating that relative pandemic impact was not always proportional to country size.
- The geographical analysis suggested that neighboring countries often experienced similar pandemic patterns, especially during major waves such as the Omicron period in late 2021 and early 2022. This may reflect regional mobility, cross-border travel, economic connectivity, and shared timing of virus transmission across Europe.
- Vaccination patterns also differed between countries. Pfizer/BioNTech (COM) was the most widely used vaccine brand across the EU/EEA, but country-level differences were still visible. These differences may be explained by vaccine availability, national vaccination policies, reporting practices, and public acceptance. The analysis of children, adolescents, and elderly populations also showed that vaccination strategies were not uniform across countries.
- The correlation analyses provided useful additional insights, but they also showed that the relationships between vaccination, deaths, cases, and healthcare burden were not always strong. Higher first-dose vaccination coverage was associated with lower COVID-19 death rates, but the relationship was only weak to moderate. Booster or additional-dose uptake did not show a strong relationship with ICU pressure, and high case rates were only weakly associated with higher ICU burden. These findings suggest that vaccination likely contributed to reducing severe outcomes, but other factors such as age structure, healthcare capacity, public health measures, virus variants, and reporting differences also played important roles.
- A limitation of this analysis is that the datasets are aggregated at country level. This means that individual-level factors, such as age, health status, previous infection, or exact vaccination timing, could not be considered. Therefore, the correlation results should be interpreted as associations, not as proof of direct causation.

11 Conclusion

- This assignment analyzed COVID-19 cases, deaths, vaccination coverage, and healthcare burden across EU/EEA countries using ECDC datasets. The findings show that COVID-19 impacts varied substantially between countries and were influenced by population size, geography, vaccination strategies, healthcare pressure, and pandemic waves.

- Overall, the analysis suggests that vaccination programs were likely important in reducing severe COVID-19 outcomes, but vaccination alone did not fully explain differences in deaths or ICU pressure across countries. The results highlight that COVID-19 was a complex public health challenge where infection spread, healthcare capacity, demographic structure, policy responses, and vaccine uptake all interacted.
- In conclusion, data analysis and visualization helped reveal important patterns in the COVID-19 pandemic across the EU/EEA. These insights can support better understanding of public health responses and may be useful for future pandemic preparedness and healthcare planning.

12 Appendix

Appendix A : Data Transformation

Table 9 Data Transformation Summary for All Datasets

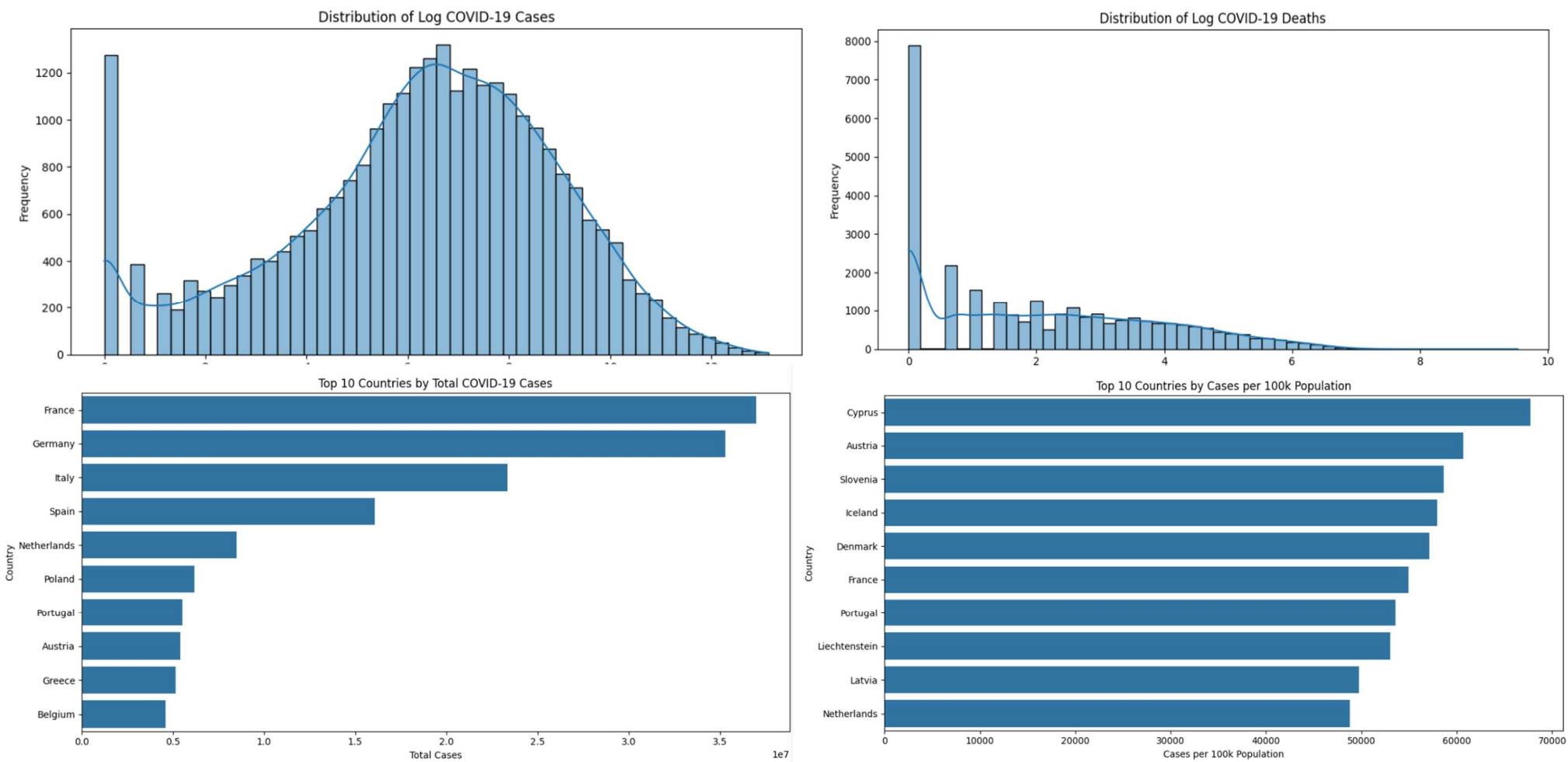
Dataset	Column	Step	Method	Motivation
Cases and deaths dataset	countriesAndTerritories, dateRep	Sorted the dataset by country and reporting date	Used sort_values() and reset the index	Ensures that the data is ordered correctly for time-based analysis
	dateRep	Created new columns: - quarter - year_quarter	1. Extracted quarter using pandas datetime functions 2. Combined year and quarter into a label such as 2020-Q1	Allows analysis by quarter to group, read, and visualize
	cases, deaths, popData2020	Created new columns: - cases_per_100k - deaths_per_100k	Divided cases/ deaths by population and multiplied by 100,000	Allows fair comparison between countries with different population sizes
	cases, deaths	Created new columns: - log_cases - log_deaths	Applied np.log1p() transformation	Reduces the effect of very large values and helps handle skewed distributions
	cases, deaths, popData2020	Created quarterly summary table	Grouped by country, year, and quarter, then calculated total cases, deaths, and population-normalized values	Supports the analysis and visualization of quarterly COVID-19 trends
Vaccination dataset	YearWeekISO	Extracted year and week and create new columns: - year - week	Used string slicing and converted values into integers	Prepares weekly vaccination data for time-based analysis
	year, week	Created column: week_start_date	Converted year-week format into a datetime value	Enables time-series visualization of vaccination progress
	FirstDose, SecondDose, Population	Created new columns: - first_dose_rate - second_dose_rate	Divided dose counts by population and multiplied by 100	Allows comparison of vaccination coverage across countries
	FirstDoseRefused, Population	Created column: - first_dose_refused_rate	Divided first-dose refusal counts by	Supports the analysis of vaccine skepticism in relation to population size

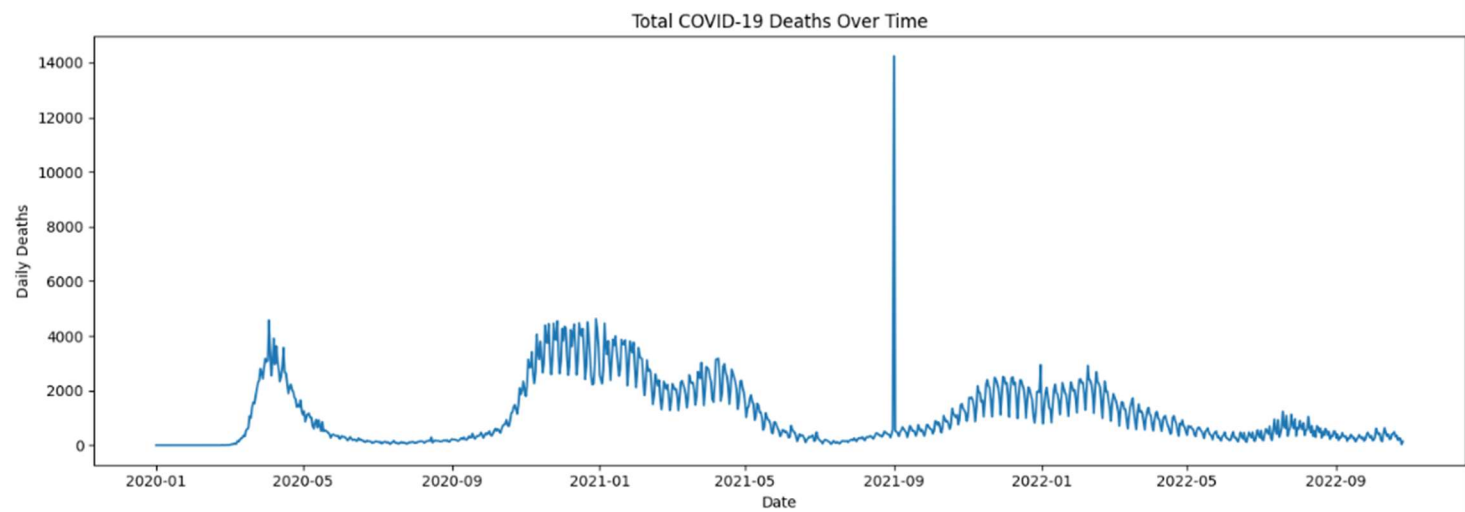
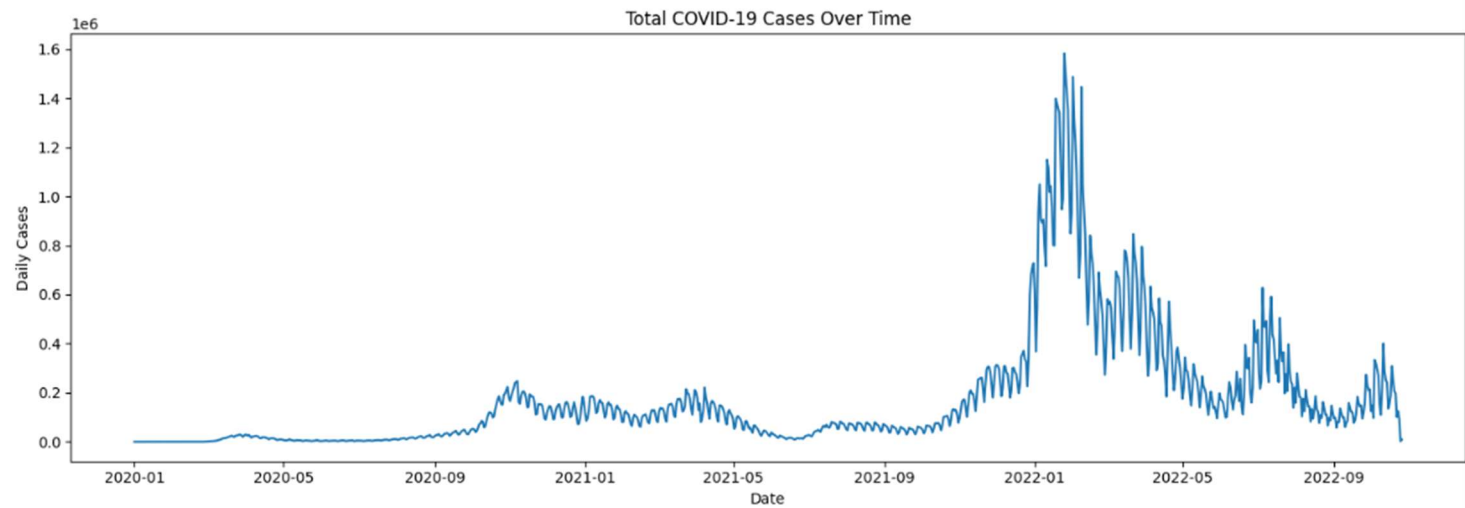
Dataset	Column	Step	Method	Motivation
			population and multiplied by 100	
	FirstDose, SecondDose	Created new columns: - log_first_dose - log_second_dose	Applied np.log1p() transformation	Reduces skewness caused by very large vaccination counts
	ReportingCountry, Vaccine	Created country-vaccine summary table	Grouped by country and vaccine brand, then aggregated first doses, second doses, received doses, exported doses, and refusals	Supports visualization and comparison of vaccine usage across EU/EEA countries
	ReportingCountry, TargetGroup, Vaccine	Created target-group vaccination summary	Grouped by country, target group, and vaccine brand	Supports analysis and visualization of vaccine distribution by target group
Hospital and ICU dataset	date	Converted date values into datetime format	Used pd.to_datetime()	Ensures that hospital and ICU data can be analyzed over time
		Extracted year and week and create new columns: - year - quarter	Extracted values using pandas datetime functions	Allows yearly and quarterly healthcare analysis
	year, quarter	Created column: year_quarter	Combined year and quarter into a label such as 2021-Q2	Makes quarterly hospital and ICU results easier to group and visualize
	country, indicator, date	Sorted the dataset by country, healthcare indicator, and date	Used sort_values() and reset the index	Keeps healthcare indicators ordered chronologically for each country
	value	Created log_value	Applied np.log1p() transformation	Reduces the impact of extreme healthcare values and skewed distributions
	country, indicator, year	Created yearly hospital/ICU summary table	Grouped by country, indicator, and year, then calculated mean, maximum, and total values	Supports comparison and visualization of healthcare impact between countries
	country, indicator, year, quarter	Created quarterly hospital/ICU summary table	Grouped by country, indicator, year, and quarter, then calculated mean,	Supports analysis and visualization of healthcare pressure across pandemic periods

Dataset	Column	Step	Method	Motivation
			maximum, and total values	

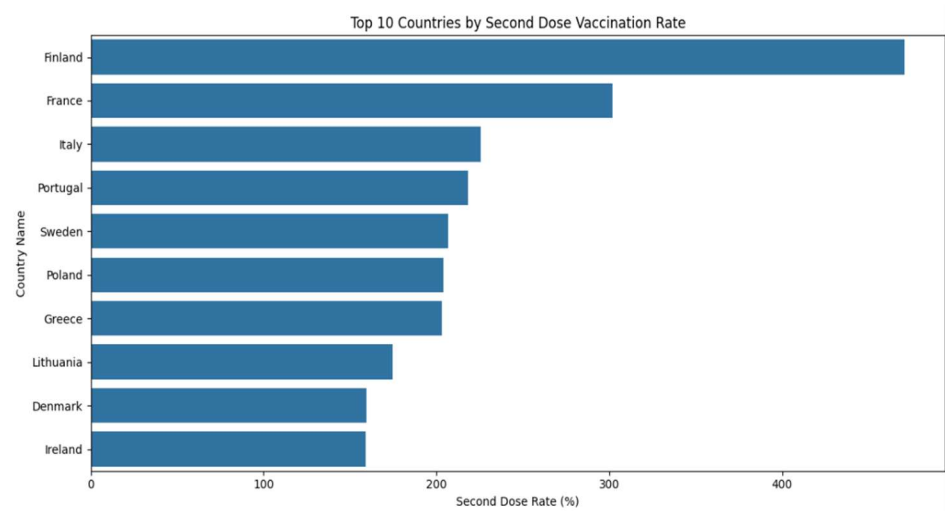
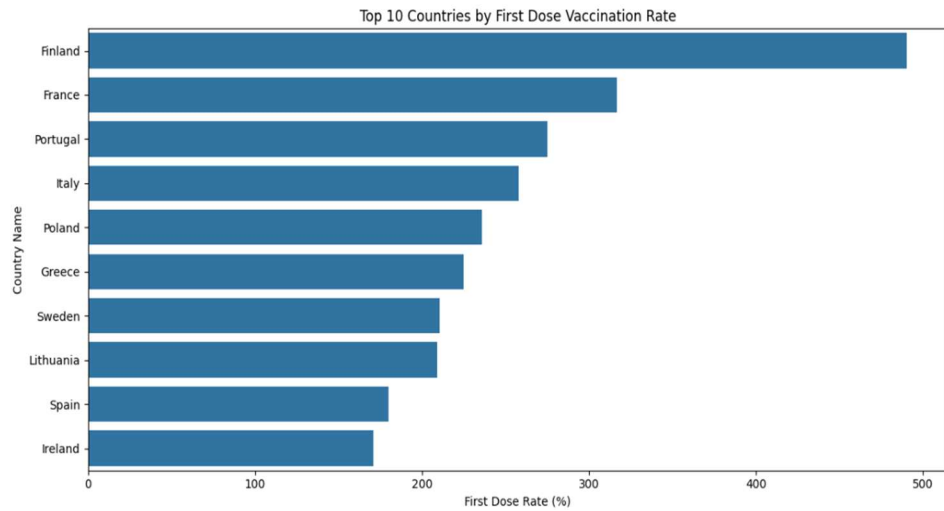
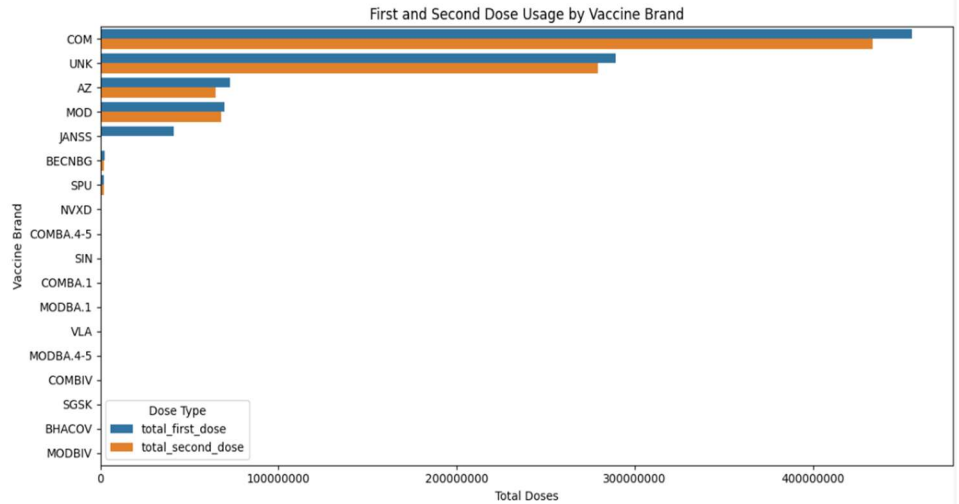
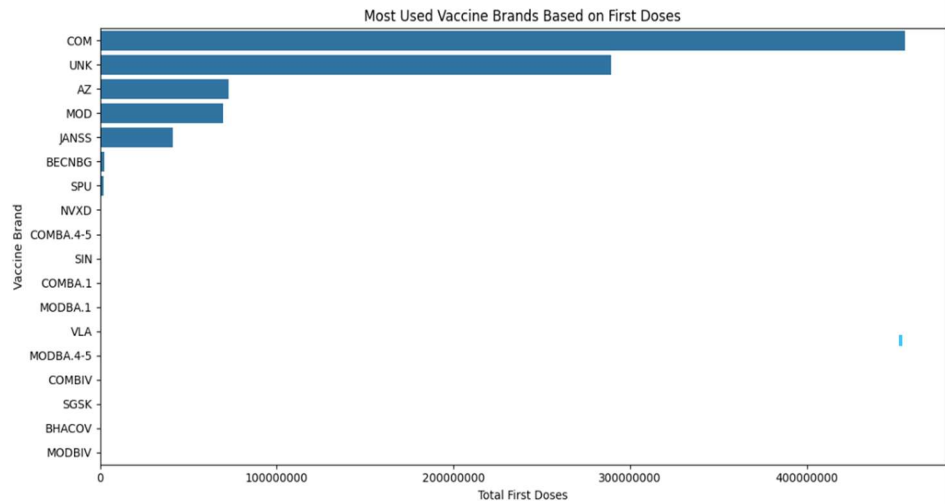
Appendix B
: Exploratory Data Analysis

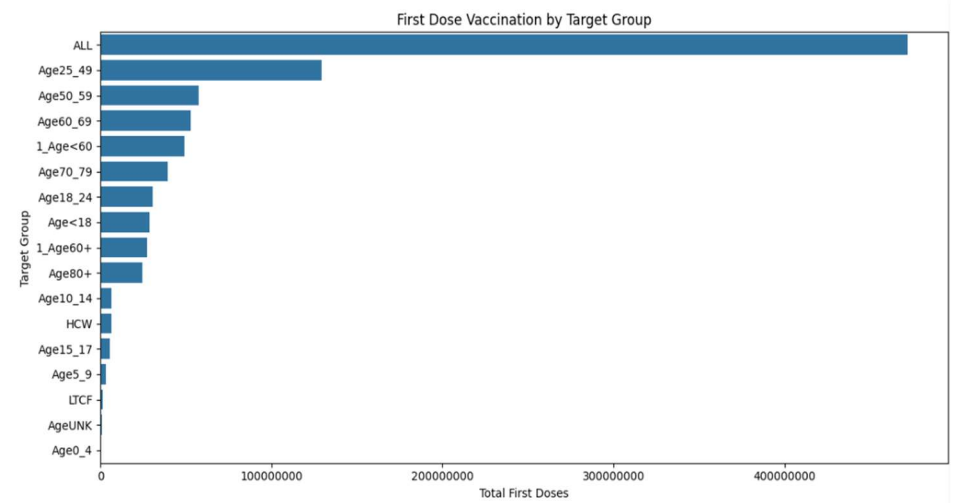
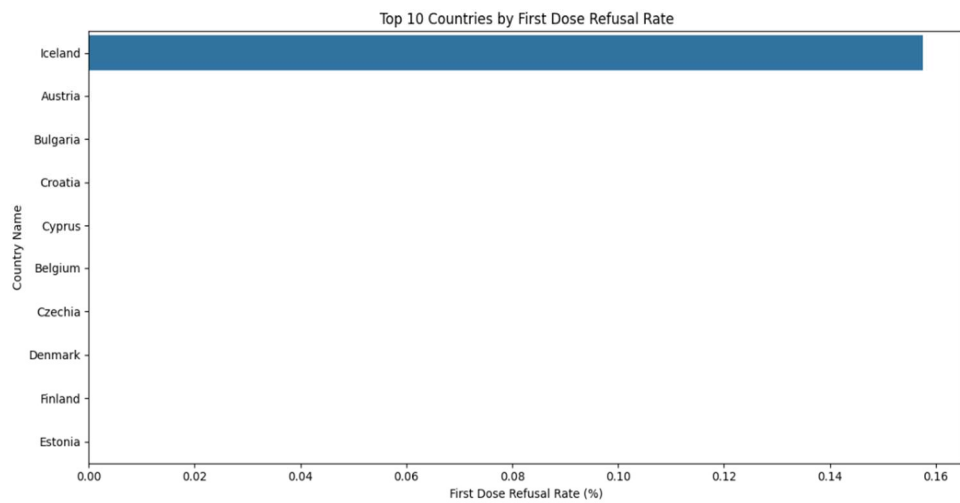
A. Cases and Death Dataset





B. Vaccination Dataset





C. Hospital and ICU Dataset

